



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch : V Semester, B.E CSE
Course Code & Title : CB3491-Cryptography and Cyber Security
Name of the Faculty member : Mrs.K.Jeyakarthika, AP / CSE

Innovative Practice Description

- **Unit / Topic:** Unit III /Primality Testing
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO 9
- **Activity Chosen:** Flipped Class Room
- **Justification:**

The flipped classroom is a teaching methodology where traditional teaching methods are reversed. Instead of introducing concepts during in-class lectures, students are introduced to the content at home (via videos, readings, or other materials), and classroom time is used for engaging activities like discussions, problem-solving, or hands-on exercises to deepen understanding. This approach promotes active learning, better engagement, and personalized instruction, as teachers can spend more time addressing individual or group challenges during class. It is commonly supported by technology to deliver pre-class materials.

- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
 - The learning materials were posted to the students in canvas two days before the topic discussion.
 - The students are instructed to study the material and take notes on the topic based on their understanding.
 - On the day of the activity the students were grouped into a team of 4 members and they are asked to discuss about the topic they learnt in their home through the study materials for the duration of 10 minutes and it is shown in Fig.1.
 - Based on the study material they have gone through, the problems were given for each group separately and by discussing with their team members they have solved it.
 - After the discussion one member from each team had a chance to present and share the way of solving the problems to the entire class as shown in Fig.2.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	3	1	2	1	3	3	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO5
	3	3	3	3	3
Justification for correlation	Apply the basic Computer Science Engineering and mathematics Knowledge for performing Symmetric encryption.	Analyze various Asymmetric encryption algorithms	Developing Asymmetric encryption systems with basic encryption mechanisms are highly correlated	Investigate various aspects of cryptanalysis is highly correlated	Use the virtual environment and crypt tool for performing symmetric encryption
	PO8	PO9	PO12		
	1	2	1		
	To complete the assignments/ tasks in an ethical way	To work effectively as an individual and team for encrypt and decrypt a message using Asymmetric Encryption Algorithms	Ability to update the Asymmetric Encryption algorithms to the current trends.		

- **Images / Screenshot of the practice:**



Figure 1: Students discussion with their team members



Fig 2. Student share their view with classmates

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- Students realized the benefits of self-learning while discussing with their group members.
 - Those who didn't get clear ideas from the study material, with the help of peer team learning they understood the concepts clearly.

- ❖ **Benefit of the practice:**

- Flipped Class room activity helps the students to remember the concepts of aggregation and composition easily
 - The students sharing their gained knowledge with their friends help them to gain more understanding about the concepts
 - This activity helps them to prepare easily during exam.
 - Understanding and remembering of the topic help the students to design any application

- ❖ **Challenges faced in implementation:**

- It is difficult to make all the students to go through the study material hence they struggle a lot to recollect the topic and not able to interact effectively among the team members.

References:

- https://en.wikipedia.org/wiki/Flipped_classroom
- https://www.ritrjpm.ac.in/images/computer-science/2021-2022/BV_CS8602_Flipped-classroom.pdf
- <https://www.ritrjpm.ac.in/images/computer-science/CS8603-FlippedClassroom.pdf>